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Fluid Loss Control by a Novel Star Polymer for Oil Based Mud

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Abstract

Non-Aqueous Fluids (NAF) are widely recognized as efficient and reliable drilling fluid systems for challenging wellbore conditions, particularly in high-temperature drilling operations. To minimize filtrate invasion and maintain a thin filter cake, NAFs require effective fluid loss control additives. The polymer developed in this study exhibits superior properties—including high dispersibility, excellent thermal stability, and minimal impact on plastic viscosity—when compared with conventional natural and synthetic fluid loss control additives such as asphaltite.

A synthetic molecular optimization approach was employed to fine-tune the hydrophilic-lipophilic balance (HLB) by adjusting the ratio of hydrophilic to hydrophobic monomers. This allowed us to achieve an optimal HLB that promotes easy dispersion within NAF formulations. The star polymer was synthesized using a controlled radical polymerization method, specifically Reversible Addition-Fragmentation Chain Transfer (RAFT) polymerization.

Key fluid properties—such as rheology, fluid loss, filter cake thickness, and emulsion stability—were evaluated in NAF systems and benchmarked against commercially available fluid loss additives under simulated downhole pressure and temperature conditions. The polymer's chemical structure and thermal stability were confirmed using spectroscopic techniques and thermogravimetric analysis (TGA).

Spectroscopic analysis verified the successful synthesis of the target polymer architecture and molecular weight. The resulting star polymer demonstrated excellent thermal stability up to 450°F, along with superior fluid loss control and formation of an ultra-thin filter cake across mud weights ranging from 10 to 17 ppg. Additionally, the polymer significantly enhanced emulsion stability. While traditional fluid loss control additives tend to increase plastic viscosity, the star polymer showed negligible impact on PV. Performance results are presented for both diesel- and mineral oil-based mud systems. Rheological properties were evaluated using a high-temperature, high-pressure viscometer (Fann 77) at temperatures up to 450°F and pressures up to 10,000 psi.

This work has resulted in the development of a novel star polymer, representing a next-generation high-performance fluid loss additive for the drilling industry. The polymeric network can be readily synthesized and scaled up for commercial production. The availability of this advanced material will enhance drilling

efficiency in high-temperature (up to 450°F) wells by reducing plastic viscosity, improving emulsion stability, and providing superior fluid loss control.

Introduction

Extraction of hydrocarbon is a very complex and expensive process, with drilling being one of the critical steps (Lukawski et al., 2014). Drilling fluid is an essential component of drilling operations, as it cools the drill bit, stabilizes the wellbore during drilling and removes the cuttings from the bottomhole to the surface. It also helps balance the subsurface pressure and contribute to the wellbore stability (Apaleke et al., 2012; Caenn and Chillingar, 1996). Drilling fluid is an optimized mixture of chemicals that can consist of solids, liquids, and gases, and can be of different types based on the base fluids used. These include water-based mud (WBM) (Dye et al., 2006; Ewy and Morton, 2009; Gbadamosi et al., 2019), non-aqueous fluids (NAF) (Adewale and Ogunrinde, 2010; Aston et al., 2002; Davies et al., 1984), and foam drilling mud (Luo et al., 2012; Negrão et al., 1999; Nugroho et al., 2017). Non-aqueous fluid (NAF) could be of diesel based, mineral oil based or synthetic oil based. Overall use of water-based and non-aqueous fluids is about 50:50 in ratio.

Non-aqueous fluid (NAF) is a type of drilling fluid used in drilling operations where water-based mud (WBM) may not be suitable due to the high temperature of the reservoir or sensitive to the formation for example swellable clay-based. There are several advantages to using Non-aqueous fluid (NAF) in drilling operations: (a) lower coefficient of friction compared to water-based mud (Growcock et al., 1999), which results in higher rates of penetration (ROP) or faster drilling, ultimately reducing drilling costs and increasing overall drilling efficiency; (b) highly effective in preventing differential sticking (Simpson, 1962) that occurs when the drill pipe becomes stuck to the formation; (c) generally stable at higher temperatures; (d) provides excellent cutting removal due to its high temperature rheological stability and (e) provides better hole stability and formation protection, especially in formations that are sensitive to water-based fluids (Jung et al., 2013; Saleh, 2022; Simpson et al., 1995).

Fluid loss control to the formation is an important aspect of drilling, wherein lower the fluid loss, lower is the filter cake thickness. A higher fluid loss can result in adversely affecting fluid rheology, on the other hand higher filter cake thickness can cause stuck pipe situations and related nonproductive time. To control fluid loss, conventional non-aqueous fluid (NAF) often incorporate additives such as Asphaltite (Guo et al., 2014; Pakdaman et al., 2020), due to its low cost. However, Asphaltite's low chemical stability can make it unsuitable for use under high-temperature and high-pressure conditions. Commercially available linear polymers (styrene-butadiene based) used non-aqueous fluid (NAF) systems (Audibert et al., 2004) do a pretty good job, however, one significant drawback is the unintended increase in viscosity due to their high molecular weight and hydrodynamic volume.

To overcome these issues, novel star polymers have been developed in this study. Star polymers are polymeric materials that have a central core with multiple polymer arms extending outward with specific properties, such as high molecular weight, high branching density, and high surface area, which make them ideal for fluid loss control in drilling fluid. One of the main advantages of star polymers is their ability to form a tight, impermeable seal at the wellbore wall, preventing fluid from entering the formation. The star polymer developed in this study demonstrated excellent fluid control for non-aqueous fluid (NAF) under different temperature conditions up to 450 °F.

Synthesis of novel star polymer (NSP)

As shown in Figure 1, monomers (A) Acrylic acid, (B) Lauryl acrylate, and (C) Methylene bisacrylamide were used for the synthesis of NSP. Controlled "living" radical polymerization technique, namely Reversible Addition-Fragmentation chain Transfer polymerization (RAFT), was employed. The stepwise synthetic route of star polymer is shown in Figure 2 wherein the chain transfer agent was synthesized in Step 1. Lauryl acrylate, and azobisisobutyronitrile (AIBN) were then added with tetrahydrofuran solvent to perform

synthesis of block copolymer poly(acrylic acid)-block-poly(lauryl acrylate) in Step 2. Thus synthesized block copolymer was then crosslinked upon reaction with methylene bisacrylamide in Step 3 in the presence azobisisobutyronitrile (AIBN) initiator to achieve a star-like architecture. An additional desulphurization process was carried out in Step 4 to produce the desired novel star polymer (NSP) as the final product free of sulfurous smell. An engineered Hydrophilic and Lyophilic Balance (HLB) ratio was obtained by properly optimizing the molecular ratio of different monomers used herein.

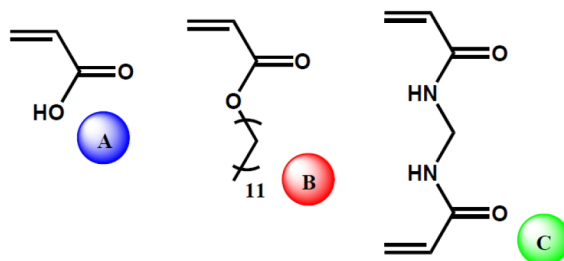


Figure 1—Monomers used for synthesis of NSP

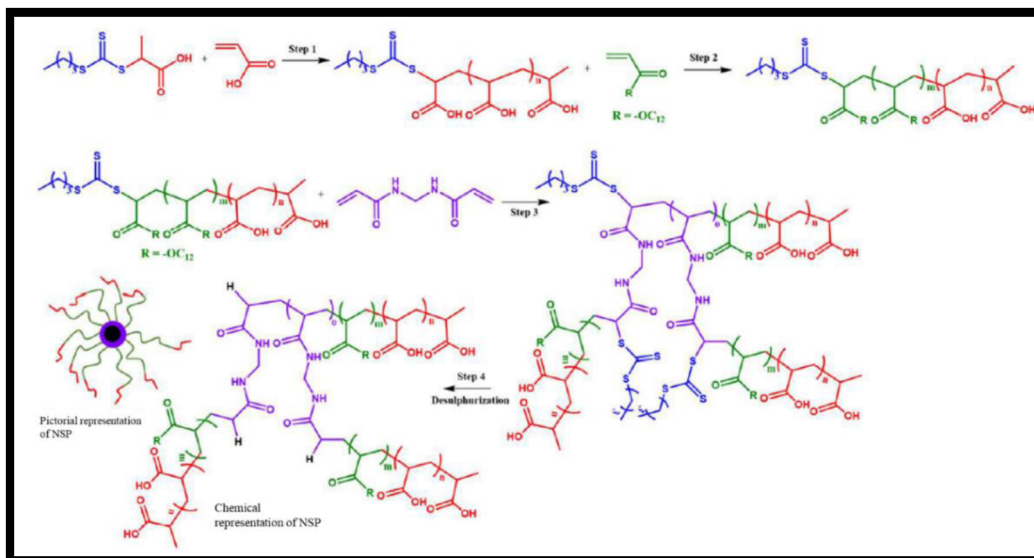


Figure 2—Synthesis of block copolymer star with hydrophilic acrylic acid and hydrophobic lauryl acrylate with methylene bisacrylamide crosslinker via controlled RAFT polymerization. Detailed synthetic scheme is described in a prior publication (Jian 2023).

Characterization of novel star polymer (NSP)

Fourier-transform infrared spectroscopy (FTIR)

Fourier-transform infrared spectroscopy (FTIR) data was collected using a Cary 630 FTIR Spectrometer (Agilent Technologies) and the processing of the data was performed on MicroLab Expert (provided by Agilent Technologies). Figure 2a shows a broad adsorption band at around 3302 cm^{-1} that can be attributed to the second amine group within the methylene bisacrylamide backbone. The 2922 and 2855 cm^{-1} absorption bands are due to C-H stretches, the 1733 cm^{-1} absorption band is due to C=O stretch within the carboxylate acid moiety, the 1457 and 1375 cm^{-1} absorption band are due to O-H stretch for the carboxylic acid group, the 1241 and 1114 cm^{-1} absorption band are due to C-N stretch within methylene bisacrylamide moiety, and the 1159 cm^{-1} absorption band is to C-O stretch within carboxylic acid group.

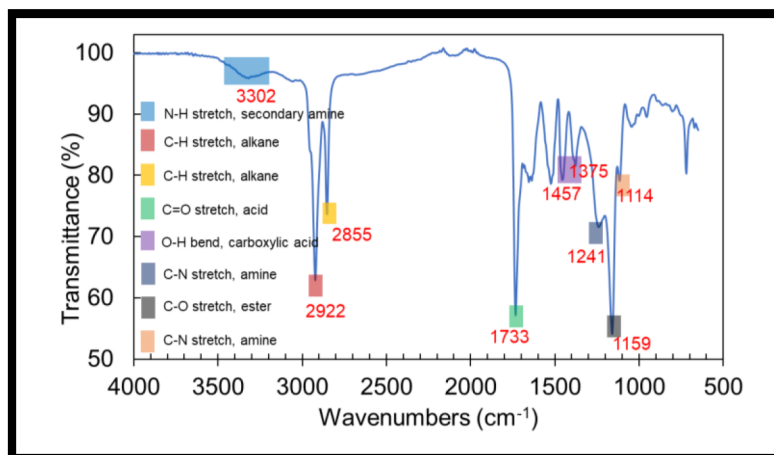


Figure 2a—FTIR spectra of NSP; Further details of the characterization are published in a prior paper (Jian 2023).

Thermal Gravimetric Analysis (TGA)

Thermogravimetric analyses (TGA) data acquisition was done on a TA Thermogravimetric Analyzer (SDT Q600). All samples tested are scanned from room temperature using a constant heating rate of 10 °C/min in air to 1000 °C.

Figure 3 shows, initial mass loss around temperature of 225°C is attributed to the loss of trapped solvent in the polymer. Heating the polymer (NSP) above 225 °C led to continuous loss of mass due to the degradation of the polymer. There are two major polymer degradation peaks at 380°C and 550 °C respectively, demonstrating the high thermal stability of NSP at least up to 380°C under ambient pressure. Higher temperature stability is expected under higher downhole pressure.

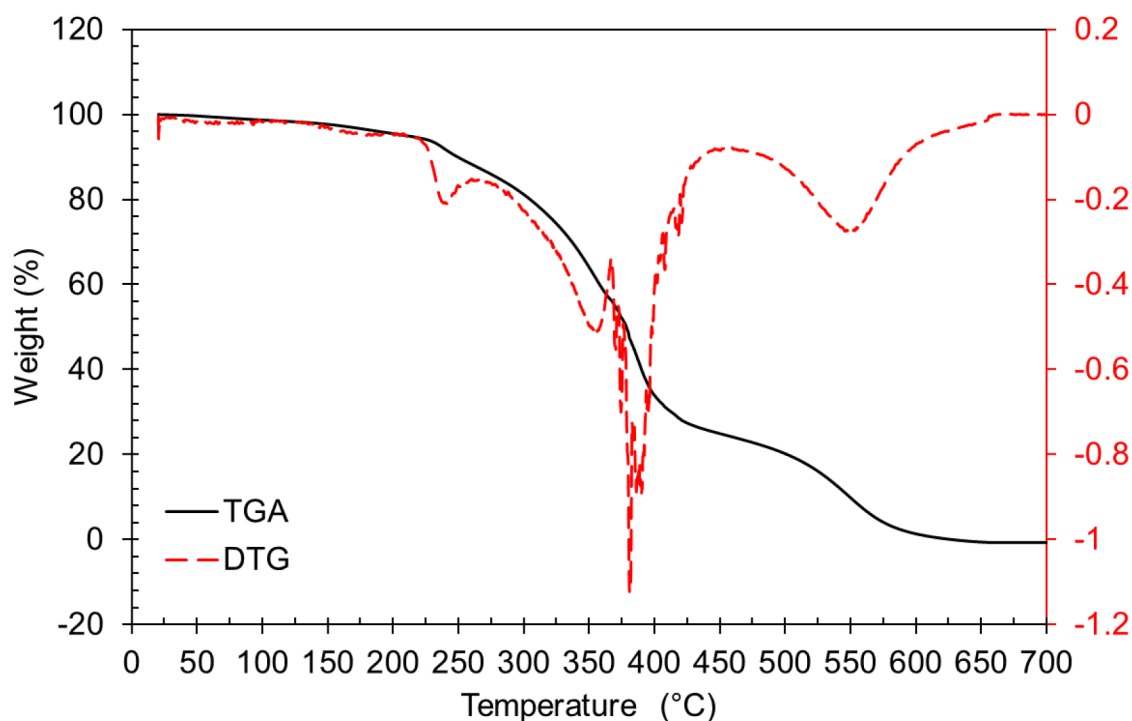


Figure 3—TGA curve for NSP; Further details of the characterization are published in a prior paper (Jian 2023).

Composition, Rheology and fluid loss of organo-clay based invert emulsion fluids

Invert emulsion-based drilling fluids were prepared using standard API method using both Diesel and Mineral oil based non-aqueous fluid (NAF).

Table 1—Non-aqueous fluid (NAF) design with a density of 13.4ppg (100pcf) using diesel

Fluid formulations (lb/bbl)	NSP based	Asphaltite based
Diesel	178.3	178.3
Organoclay	4	4
Rheology modifier	2	2
Lime	10	10
Primary emulsifier	10	10
Wetting agent	5	5
CaCl ₂ brine	71.85	71.85
NSP	3.6	-
Asphaltite based	-	4
Barite	280	280
RevDust	50	50
Aging temp., °F	300	300
Aging period, hr	16	16

Table 2—Non-aqueous fluid (NAF) with a density of 13.4ppg (100pcf) using Mineral oil

Fluid formulations (lb/bbl)	NSP	Asphaltite based
Mineral oil	157.6	157.6
Organoclay	4	4
Rheology modifier	2	2
Lime	10	10
Primary emulsifier	10	10
Wetting agent	5	5
CaCl ₂ brine	63.27	63.27
NSP	4	-
Asphaltite based	-	4
Barite	310	310
RevDust	50	50
Aging temp., °F	300	300
Aging period, hr	16	16

Rheology results in [Table 3](#) show that the diesel based invert emulsion fluid with NSP significantly outperforms the Asphaltite based one. NSP base fluid has higher 3 and 6RPM values, lower PV, higher YP and non-progressive gel strength development, all these are excellent properties of an invert emulsion fluid contributing to higher drill cutting carrying capacity, lower equivalent circulating density (ECD) and superior hole cleaning capacity.

Table 3—Mud rheology measurement of 13.4 ppg (100pcf) diesel based NAF at 150 °F

Fann35	NSP	Asphaltite based
600 rpm	96	130
300 rpm	57	72
200 rpm	46	52
100 rpm	30	30
6 rpm	12	4
3 rpm	11	3
PV, cp	39	58
YP, lb/100 ft ²	18	14
10 sec. gel, lb/100 ft ²	16	6
10 min. gel, lb/100 ft ²	35	40

Greater emulsion stability, lower filter cake thickness and lower fluid loss have been demonstrated in [Table 4](#) for the diesel based fluid with NSP, indicating properties of an effective drilling fluid.

Table 4—Fluid loss measurement of 13.4 ppg NAF with diesel at 350°F

Fluid loss control additive	Fluid loss volume (mL)	Filter cake thickness	ES (V)
NSP	<1	2/32"	550
Asphaltite based	8.2	16/32"	255

Rheology results in [Table 5](#) show that the mineral oil based invert emulsion fluid with NSP significantly outperforms the Asphaltite based one. NSP base fluid has lower PV, higher YP and non-progressive gel strength development, all these are excellent properties of an invert emulsion fluid contributing to higher drill cutting carrying capacity, lower equivalent circulating density (ECD) and superior hole cleaning capacity.

Table 5—Mud rheology measurement of 13.4 ppg Mineral oil based NAF at 150 °F

Fann 35	NSP	Asphaltite based
600 rpm	112	125
300 rpm	68	73
200 rpm	51	55
100 rpm	33	34
6 rpm	14	13
3 rpm	12	10
PV, cp	44	52
YP, lb/100 ft ²	24	21
10 sec. gel, lb/100 ft ²	31	24
10 min. gel, lb/100 ft ²	42	57

Greater emulsion stability, lower filter cake thickness and lower fluid loss have been demonstrated in [Table 6](#) for the mineral oil based fluid with NSP, indicating properties of an effective drilling fluid.

Table 6—Fluid loss measurement of 13.4 ppg Mineral oil based OBM at 350°F

Fluid loss control additive	Fluid loss volume (mL)	Filter cake thickness	ES (V)
NSP	2.8	6/32"	470
Asphaltite based	2.6	22/32"	350

Composition, Rheology and fluid loss of organo-clay free invert emulsion fluids

Invert emulsion-based drilling fluids were prepared using standard API method using both Diesel and Mineral oil without the use of organo-clay to improve formation damage properties.

Table 7—NAF designs at density of 13.4ppg (100pcf) using Diesel and Mineral oil

Fluid formulations (lb/bbl)	Diesel based	Mineral oil based
Mineral oil	-	157.6
Diesel	178.3	-
Nanomaterial based viscosifier	2	2
Rheology modifier	3	3
Lime	10	10
Primary emulsifier	10	10
Secondary emulsifier	5	5
Wetting agent	2	2
CaCl ₂ brine	85	85
NSP	4	4
Barite	280	280
RevDust	50	50
Aging temp., °F	300	300
Aging period, hr	16	16

Rheology results in [Table 8](#) show that the both clay-free diesel and mineral oil based invert emulsion fluids with NSP have higher 3 and 6RPM values, lower PV, higher YP, all these are excellent properties of an invert emulsion fluid contributing to higher drill cutting carrying capacity, lower equivalent circulating density (ECD) and superior hole cleaning capacity. Both fluids show excellent emulsion stability, fluid low control and filter cake thickness ([Table 9](#)).

Table 8—Fann 35 rheology measured at 150 °F after aging at 300 °F

Dial Reading	Diesel based	Mineral oil based
600	117	113
300	71	73
200	54	57
100	36	40
6	13	19
3	12	17
10s gel	15	22
10m gel	23	32
PV, cp	46	40
YP, lb/100 ft ²	25	33

Table 9—Fluid Loss Characteristics of NAF fluids at 350 °F

Drilling Fluids	Fluid loss, mL	Cake thickness at 350 °F	ES (V)
Diesel based	1.6	2/32"	696
Mineral oil based	1.6	2/32"	670

The high temperature rheological stability of clay-free NAF under extreme conditions, 10000 psi and up to 450 °F, was also studied through Fann 77 rheometer (Table 10 & 11). The results clearly demonstrate that the 3 and 6RPM readings, YP are pretty stable until up to 450 °F indicative of a very healthy fluid even at that higher temperature.

Table 10—Fann 77 rheology clay-free diesel based NAF at 10000 psi and up to 450 °F.

RPM	Dial Reading						
	150 °F	200 °F	250 °F	300 °F	350 °F	400 °F	450 °F
600	130	94	65	45	40	33	34
300	82	62	39	28	25	22	23
200	69	45	31	18	15	12	14
100	44	29	21	12	10	9	11
6	14	11	11	8	6	6	8
3	13	10	10	6	6	5	6
PV, cp	48	32	26	17	15	11	11
YP, lb/100 ft ²	34	30	13	11	10	11	12

Table 11—Fann 77 rheology clay-free mineral oil based NAF at 10000 psi and up to 450 °F.

RPM	Dial Reading						
	150 °F	200 °F	250 °F	300 °F	350 °F	400 °F	450 °F
600	130	80	62	46	42	41	42
300	80	54	39	28	26	26	26
200	62	42	30	21	18	19	18
100	41	28	22	15	12	12	11
6	16	13	12	8	7	5	5
3	15	12	11	7	6	5	5
PV, cp	50	26	23	18	16	15	16
YP, lb/100 ft ²	30	28	16	10	10	11	10

Conclusions

- Characterization results of NSP demonstrate that it is thermally stable up to 380°C (716°F) at ambient temperature, as indicated by TGA. In addition, FTIR confirmed the presence of the desired functional groups in the synthesized polymer.
- NSP could produce *organo-clay based* non-aqueous invert emulsion-based fluids both in diesel and mineral oil-based system with higher emulsion stability, higher YP, lower ECD, lower fluid loss and filter cake thickness.
- NSP could produce *clay-free* non-aqueous invert emulsion-based fluids both in diesel and mineral oil-based system with higher emulsion stability, higher YP, lower ECD, lower fluid loss and filter cake thickness.
- NSP based fluids demonstrated thermal stability all the way up to 450 °F as demonstrated by their rheological properties measured by Fann 77, a HTHP rheometer.

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